
EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 8A

For this discussion, **Note 8** is helpful.

1. Proctoring Practice

To prepare for the midterm coming up on **March 15, 2021**, let's take a minute to ensure that the proctoring system will work.

Find the email we sent you with a Zoom link (it should have subject line "[EECS 16B] Personal Zoom Proctoring Link for Exams"), join this meeting, and record yourself for five minutes.

In case something went wrong with your Zoom room, fill out **this form** to tell us what went wrong so we can fix it for you.

2. BIBO Stability

Consider a continuous-time scalar real differential equation with known solution

$$\frac{d}{dt}x(t) = ax(t) + bu(t) \quad x(t) = e^{at}x(0) + \int_0^t e^{a(t-\tau)}bu(\tau) d\tau.$$

Show that if the system has $\operatorname{Re}\{a\} > 0$, then a bounded input can result in an unbounded output (i.e., the system is unstable) for every initial condition $x(0)$.

3. Changing behavior through feedback

In this question, we discuss how feedback control can be used to change the effective behavior of a system.

(a) Consider the scalar system:

$$x[t+1] = 0.9x[t] + u[t] + w[t] \quad (1)$$

where $u[t]$ is the control input we get to apply based on the current state and $w[t]$ is the external disturbance, each at time t .

Is the system stable? If $|w[t]| \leq \epsilon$, what can you say about $|x[t]|$ at all times t if you further assume that $u[t] = 0$ and the initial condition $x[0] = 0$? How big can $|x[t]|$ get?

(b) Suppose that we decide to choose a control law $u[t] = kx[t]$ to apply in feedback. For what values of λ can you get the system to behave like:

$$x[t+1] = \lambda x[t] + w[t]? \quad (2)$$

How would you pick k ?

(Note: In this case, $w[t]$ can be thought of like another input to the system, except we can't control it.)

(Note: In lecture we call this term f – for feedback – instead of k , but we use k here since it's a more traditional notation for feedback, and also lowercase f is confused with functions.)

(c) For the previous part, which k would you choose to minimize how big $|x[t]|$ can get?

(d) What if instead of a 0.9, we had a 3 in the original eq. (1). What, if anything, would change?

(e) Now suppose that we have a vector-valued system with a vector-valued control:

$$\vec{x}[t+1] = A\vec{x}[t] + B\vec{u}[t] + \vec{w}[t] \quad (3)$$

where we further assume that B is an invertible square matrix.

Suppose we decide to apply linear feedback control using a square matrix F so we choose $\vec{u}[t] = F\vec{x}[t]$.

For what values of matrix G can you get the system to behave like:

$$\vec{x}[t+1] = G\vec{x}[t] + \vec{w}[t] \quad (4)$$

How would you pick F given knowledge of A, B and the desired goal dynamics G ?

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